

Hints & Solutions

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Q1 (4) - Single Correct

We have, $\sqrt{5x - 6 - x^2} + \frac{\pi}{2} \int_0^x dz > x \int_0^\pi \sin^2 x dx$

$$\Rightarrow \sqrt{5x - 6 - x^2} + \frac{\pi}{2} x > \frac{\pi}{2} x$$

$$\Rightarrow \sqrt{5x - 6 - x^2} > 0$$

$$\Rightarrow 5x - 6 - x^2 > 0$$

$$\Rightarrow x^2 - 5x + 6 < 0$$

$$\Rightarrow (x - 2)(x - 3) < 0 \Rightarrow 2 < x < 3$$

$$\therefore x \in (2, 3)$$

Q2 (3) - Single Correct

Given, $U_n = \int_0^1 x^n (2 - x)^n dx$

and $V_n = \int_0^1 x^n (1 - x)^n dx$

In U_n , put $x = 2t \Rightarrow dx = 2dt$

$$\therefore U_n = \int_0^{1/2} 2^n \cdot t^n \cdot (2 - 2t)^n \cdot 2dt \dots (i)$$

$$\text{Now, } V_n = 2 \int_0^{1/2} x^n \cdot (1 - x)^n dx \dots (ii)$$

From Eqs. (i) and (ii), we get

$$U_n = 2^{2n} \cdot V_n$$

Q3 (1) - Single Correct

Given, $\int_0^{f(x)} t^2 dt = x \cos \pi x$

On differentiating w.r.t. x both sides, we get

$$[f(x)]^2 f'(x) = \pi x \sin \pi x + \cos \pi x$$

On putting $x = 9$, we get

$$[f(9)]^2 f'(9) = -1 \dots (i)$$

$$\text{Also, } \left[\frac{t^3}{3} \right]_0^{f(x)} = x \cos \pi x$$

$$\Rightarrow \frac{[f(x)]^3}{3} - 0 = x \cos \pi x$$

$$\Rightarrow [f(9)]^3 = 3 \cdot 9 \cos 9\pi = -27$$

$\therefore f(9) = -3$, putting in Eq. (i), we get

$$(-3)^2 f'(9) = -1$$

$$\Rightarrow f'(9) = -\frac{1}{9}$$

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Q4 (1) - Single Correct

$$\text{Let } f(a) = \int_a^{a^2} \frac{dx}{x + \sqrt{x}}$$

$$\therefore f'(a) = \frac{1}{a^2 + a} \times 2a - \frac{1}{a + \sqrt{a}}$$

$$\Rightarrow \frac{2}{a+1} - \frac{1}{a+\sqrt{a}} = 0$$

$$\Rightarrow 2a + 2\sqrt{a} - a - 1 = 0$$

$$\Rightarrow 2\sqrt{a} + a - 1 = 0$$

$$\Rightarrow 2\sqrt{a} = -a + 1$$

$$\Rightarrow 4a = a^2 + 1 - 2a$$

$$\Rightarrow a^2 - 2a - 4a + 1 = 0$$

$$\Rightarrow a^2 - 6a + 1 = 0$$

$$\Rightarrow a = \frac{6 \pm \sqrt{36-4}}{2} = \frac{6 \pm 4\sqrt{2}}{2}$$

$$\Rightarrow a = 3 \pm 2\sqrt{2} = (\sqrt{2} \pm 1)^2$$

$$\text{and } f''(a) = \frac{2}{2\sqrt{a}} + 1$$

$$\text{It is smallest, when } \sqrt{a} = \tan \pi/8$$

$$\therefore a = \tan^2 \pi/8$$

Q5 (4) - Single Correct

$$\text{We have, } f''(x) = -\frac{1}{x^2} - \pi^2 \sin \pi x$$

$$\Rightarrow f'(x) = -\frac{x^{-1}}{-1} + \pi^2 \frac{\cos \pi x}{\pi} + C_1$$

$$\Rightarrow f'(x) = \frac{1}{x} + \pi \cos \pi x + C_1$$

$$\Rightarrow f(x) = \log x + \pi \frac{\sin \pi x}{\pi} + C_1 x + C_2$$

$$\Rightarrow f(x) = \log x + \sin \pi x + C_1 x + C_2$$

$$\Rightarrow f(1) = C_1 + C_2 = 0$$

$$\Rightarrow f'(2) = \frac{1}{2} + \pi \cos 2\pi + C_1$$

$$\Rightarrow \pi + \frac{1}{2} = \frac{1}{2} + \pi + C_1 \quad \left[\text{given, } f'(2) = \pi + \frac{1}{2} \right]$$

$$\Rightarrow C_1 = 0$$

On putting $C_1 = 0$ in Eq. (i), we get

$$0 + C_2 = 0$$

$$\Rightarrow C_2 = 0$$

$$\therefore f(x) = \log x + \sin \pi x$$

$$\Rightarrow f\left(\frac{1}{2}\right) = \log \frac{1}{2} - \sin \frac{\pi}{2} = -\log 2 + 1$$

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Q6 (1) - Single Correct

We have, $I(a) = \int_0^\pi \left(\frac{x}{a} + a \sin x \right)^2 dx$

$$\Rightarrow I(a) = \int_0^\pi \left(\frac{x^2}{a^2} + a^2 \sin^2 x + 2x \sin x \right) dx$$

$$\Rightarrow I(a) = \int_0^\pi \left(\frac{x^2}{a^2} + \frac{a^2(1-\cos 2x)}{2} + 2x \sin x \right) dx$$

$$= \left[\frac{x^3}{3a^2} + \frac{a^2}{2} \left(x - \frac{1}{2} \sin 2x \right) + 2(-x \cos x + \sin x) \right]_0^\pi$$

$$= \frac{\pi^3}{3a^2} + \frac{a^2}{2}(\pi) + 2(-\pi \cos \pi + 0)$$

$$\Rightarrow I(a) = \frac{\pi^3}{3a^2} + \frac{\pi a^2}{2} + 2\pi$$

$$I'(a) = \frac{\pi^3}{3} \left(\frac{-2}{a^3} \right) + \pi a = 0$$

$$\Rightarrow -\frac{2}{3} \frac{\pi^3}{a^3} = -\pi a$$

$$\Rightarrow \frac{2}{3} \pi^2 = a^4$$

$$\Rightarrow a = \pm \sqrt{\pi \sqrt{\frac{2}{3}}}$$

$$\Rightarrow I''(a) = -\frac{2}{3} \pi^3 \frac{(-3)}{a^4} + \pi = \frac{2\pi^3}{a^4} + \pi$$

for $a = \sqrt{\pi \sqrt{\frac{2}{3}}}$, which is minima.

Q7 (4) - Single Correct

We have, $I = \int_{\pi/2}^{3\pi/2} e^{-x/2} \frac{\sqrt{1-\sin x}}{1+\cos x} dx$

$$= \int_{\pi/2}^{3\pi/2} e^{-x/2} \left(\frac{\cos \frac{x}{2} - \sin \frac{x}{2}}{2 \cos^2 \frac{x}{2}} \right) dx$$

$$= \frac{1}{2} \int_{\pi/2}^{3\pi/2} e^{-x/2} \left[\sec \frac{x}{2} - \sec \frac{x}{2} \tan \frac{x}{2} \right] dx$$

Put $x = -2t \Rightarrow dx = -2dt$

$$\therefore I = \frac{1}{2}(-2) \int_{-\pi/4}^{-3\pi/4} e^t [\sec t + \sec t \tan t] dt$$

$$= \int_{-3\pi/4}^{-\pi/4} e^t [\sec t + \sec t \tan t] dt$$

$$= [e^t \sec t]_{-3\pi/4}^{-\pi/4} = e^{-\pi/4} \sqrt{2} + e^{-3\pi/4} \sqrt{2}$$

$$= \sqrt{2} [e^{-\pi/4} + e^{-3\pi/4}]$$

Q8 (3) - Single Correct

We have, $f(x) = \int_2^x \frac{dt}{\sqrt{1+t^4}}$

$$f'(x) = \frac{1}{\sqrt{1+x^4}}$$

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and $\text{gof}(x) = x$

Now, on differentiating w.r.t.x, we get

$$g' \circ f(x) \cdot f'(x) = 1$$

On putting $x = 2$, we get

$$g' \circ f(2) = \frac{1}{f'(2)} = \sqrt{17}$$

$$f(2) = 0$$

$$\text{Hence, } g'(0) = \sqrt{17}$$

Q9 (1) - Single Correct

We have, $a_n = \int_0^{\pi/2} (1 - \sin t)^n \sin 2t dt$

$$a_n = \int_0^{\pi/2} (1 - \sin t)^n 2 \sin t \cos t dt$$

$$\text{Put } 1 - \sin t = u$$

$$\Rightarrow \cos t dt = -du$$

$$\therefore a_n = - \int_1^0 2u^n (1 - u) du = 2 \int_0^1 (u^n - u^{n+1}) du$$

$$= 2 \left[\frac{u^{n+1}}{n+1} - \frac{u^{n+2}}{n+2} \right]_0^1 = 2 \left[\frac{1}{n+1} - \frac{1}{n+2} \right]$$

$$\text{Therefore, } \frac{a_n}{n} = 2 \left[\frac{1}{n(n+1)} - \frac{1}{n(n+2)} \right]$$

$$\therefore \lim_{n \rightarrow \infty} \sum_{n=1}^n \frac{a_n}{n} = 2 \left[\sum \left(\frac{1}{n} - \frac{1}{n+1} \right) - \frac{1}{2} \sum \left(\frac{1}{n} - \frac{1}{n+2} \right) \right]$$

$$= 2 \sum_{n=1}^n \left(\frac{1}{n} - \frac{1}{n+1} \right) - \sum_{n=1}^n \left(\frac{1}{n} - \frac{1}{n+2} \right)$$

$$= 2(1) - \left[\left(1 - \frac{1}{3} \right) + \left(\frac{1}{2} - \frac{1}{4} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \dots \right]$$

$$= 2 - \frac{3}{2} = \frac{1}{2}$$

Q10 (4) - Single Correct

We have, $\lim_{n \rightarrow \infty} n^2 \cdot C_n$ [it is $0 \cdot \infty$ form]

$$= \lim_{n \rightarrow \infty} \frac{C_n}{\frac{1}{n^2}} \left[\frac{0}{0} \text{ form} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{C'_n}{\frac{-2}{n^3}} \dots (i)$$

Now, we have $C_n = \int_{1/(n+1)}^{1/n} \frac{\tan^{-1} nx}{\sin^{-1} nx} dx$

$$C'_n = \frac{\tan^{-1} 1}{\sin^{-1} 1} \cdot \left(-\frac{1}{n^2} \right) - \frac{\tan^{-1} \left(\frac{n}{n+1} \right)}{\sin^{-1} \left(\frac{n}{n+1} \right)} \cdot \left(-\frac{1}{(n+1)^2} \right)$$

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$$\Rightarrow C'_n = -\frac{1}{2n^2} + \frac{1}{(n+1)^2} \cdot \frac{\tan^{-1}\left(\frac{n}{n+1}\right)}{\sin^{-1}\left(\frac{n}{n+1}\right)}$$

On putting Eq. (i), we get

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \frac{-\frac{1}{2n^2} + \frac{1}{(n+1)^2} \cdot \frac{\tan^{-1}\left(\frac{n}{n+1}\right)}{\sin^{-1}\left(\frac{n}{n+1}\right)}}{-\frac{1}{2n^2} + \frac{1}{2(n+1)^2}} \\ &= \lim_{n \rightarrow \infty} \frac{-\frac{2}{n^3} \left[\frac{1}{(n+1)^2} - \frac{1}{n^2} \right]}{-\frac{2}{n^3} \left[\frac{n^2 - n^2 - 1 - 2n}{n^2(n+1)^2} \right]} \\ &= \lim_{n \rightarrow \infty} \frac{2n+1}{2n^2(n+1)^2} \times \frac{n^3}{2} \\ &= \frac{1}{4} \lim_{n \rightarrow \infty} \frac{2n^4 + n^3}{n^2(n+1)^2} = \frac{1}{4} \times 2 = \frac{1}{2} \end{aligned}$$

Q11 (1) - Single Correct

We have, $g(x) = \int_2^x \frac{t}{1+t^4} dt$

Put $t^2 = u$

$$\Rightarrow t dt = \frac{1}{2} du$$

$$\therefore g(x) = \frac{1}{2} \int_4^{x^2} \frac{du}{1+u^2} = \frac{1}{2} [\tan^{-1} u]_4^{x^2}$$

$$\Rightarrow g(x) = \frac{1}{2} [\tan^{-1} x^2 - \tan^{-1} 4]$$

$$\therefore g(2) = 0$$

We have, $f(x) = e^{g(x)}$

$$f'(x) = e^{g(x)} \cdot g'(x)$$

and $g'(x) = \frac{x}{1+x^4}$

$$\Rightarrow f'(x) = e^{g(x)} \cdot \frac{x}{1+x^4}$$

$$\Rightarrow f'(2) = e^{g(2)} \cdot \frac{2}{17} = e^0 \cdot \frac{2}{17}$$

$$\Rightarrow f'(2) = \frac{2}{17}$$

Q12 (2) - Single Correct

We have,

$$f(x) = \sin x + \int_0^x f'(t) (2 \sin t - \sin^2 t) dt$$

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On differentiating both sides, we get

$$\begin{aligned} f'(x) &= \cos x + f'(x) (2 \sin x - \sin^2 x) \\ \Rightarrow f'(x) [1 - 2 \sin x + \sin^2 x] &= \cos x \\ \Rightarrow f'(x) &= \frac{\cos x}{(\sin x - 1)^2} \end{aligned}$$

$$\text{So, } f(x) = \int \frac{\cos x}{(\sin x - 1)^2} dx$$

Put $\sin x - 1 = t$

$$\begin{aligned} \Rightarrow \cos x dx &= dt \\ \therefore f(x) &= \int \frac{dt}{t^2} = -\frac{1}{t} + C \\ \Rightarrow f(x) &= -\frac{1}{\sin x - 1} + C \end{aligned}$$

On putting $x = 0$, then

$$f(0) = 1 + C \Rightarrow 0 = 1 + C$$

$$\begin{aligned} \text{Hence, } C &= -1 \\ \therefore f(x) &= \frac{1}{1 - \sin x} - 1 = \frac{1 - 1 + \sin x}{1 - \sin x} \\ &= \frac{\sin x}{1 - \sin x} \end{aligned}$$

Q13 (2) - Single Correct

We have, $\int_{-1}^1 (px + q) (x^{2n+1} + a_n x + b_n) dx = 0$

$$\Rightarrow \int_{-1}^1 [px^{2n+2} + pa_n x^2 + b_n px + qx^{2n+1} + a_n qx + b_n q] dx = 0$$

$$\Rightarrow \int_{-1}^1 [px^{2n+2} + p \cdot a_n x^2 + x(b_n p + a_n q) + qx^{2n+1} + b_n q] dx = 0$$

Since, term having odd exponent have the integral zero. $\therefore \int_{-1}^1 (px^{2n+2} + a_n px^2 + qb_n) dx$

$$= 2 \int_0^1 (px^{2n+2} + a_n px^2 + qb_n) dx$$

$$= 2 \left[p \frac{x^{2n+3}}{2n+3} + a_n \cdot p \frac{x^3}{3} + b_n q \cdot x \right]_0^1$$

$$= 2 \left[\frac{p}{2n+3} + \frac{a_n p}{3} + b_n q \right] = 0$$

$$\text{Hence, } b_n = 0 \text{ and } a_n = -\frac{3}{2n+3}$$

Q14 (3) - Single Correct

We have, $f(x) = \int_{-1}^x e^{t^2} dt$

and $h(x) = f[1 + g(x)]; g(x) < 0$ for $x > 0$

$$\text{Now } h(x) = \int_{-1}^{1+g(x)} e^{t^2} dt = f[1 + g(x)]$$

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$$\Rightarrow h'(x) = e^{\{1+g(x)\}^2} \cdot g'(x)$$

$$\Rightarrow h'(1) = e$$

$$\Rightarrow e^{\{1+g(1)\}^2} \cdot g'(1) = e$$

$$[1 + g(1)]^2 = 1$$

$$1 + g(1) = \pm 1$$

$$\therefore g(1) = 0 \text{ [not possible]}$$

$$\text{Or } g(1) = -2$$

Q15 (3) - Single Correct

$$\text{We have, } \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2 x^2}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{1 + \frac{k^2}{n^2} x^2}$$

$$= \int_0^1 \frac{1}{1 + \theta^2 x^2} d\theta$$

$$= \frac{1}{x} [\tan^{-1} \theta x]_0^1$$

$$= \frac{1}{x} [\tan^{-1} x] = \frac{\tan^{-1} x}{x}$$

Q16 (4) - Multiple Correct

$$\text{We have, } \int_0^a f(x) dx + \int_0^b f^{-1}(x) dx$$

$$\text{Now, } \int_0^b f^{-1}(x) dx$$

$$\text{Put } f^{-1}(x) = t \Rightarrow x = f(t)$$

$$\Rightarrow dx = f'(t) dt$$

$$\int_0^{b-1} f(x) dx = \int_0^a t f'(t) dt = [t \cdot f(t)]_0^a - \int_0^a f(t) dt$$

$$= af(a) - \int_0^a f(t) dt$$

On putting in Eq. (i), we get

$$\int_0^a f(x) dx + af(a) - \int_0^a f(t) dt = af(a) = ab$$

Q17 (1, 3) - Multiple Correct

$$\text{We have, } \int_1^{xy} f(t) dt = y \int_1^x f(t) dt + x \int_1^y f(t) dt$$

On differentiating w.r.t. x , treating y constant, we get

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$$f(xy) \cdot y = yf(x) + \int_1^y f(t)dt$$

On putting $x = 1$, we get

$$f(y)y = yf(1) + \int_1^y f(t)dt$$

$$\Rightarrow yf(y) = 3y + \int_1^y f(t)dt$$

On differentiating w.r.t. y , we get

$$yf'(y) + f(y) = 3 + f(y)$$

$$\Rightarrow f'(y) = \frac{3}{y}$$

$$\Rightarrow f(y) = 3 \log y + C$$

On putting $y = 1$, we get

$$f(1) = 3 = C$$

$$\text{NOW, } f(y) = 3 \log y + 3 = 3(\log y + 1)$$

$$\therefore f(y) = 3 \log(ey)$$

$$\text{Hence, } f(x) = 3 \log ex$$

Q18 (3) - Multiple Correct

$$\text{Since, } x \in \left[0, \frac{\pi}{2}\right]$$

$$\Rightarrow 1 \leq 1 + \sin^3 x \leq 2$$

$$\Rightarrow \frac{1}{\sqrt{2}} \leq \frac{1}{\sqrt{1+\sin^3 x}} \leq 1$$

$$\Rightarrow \int_0^{\pi/2} \frac{1}{\sqrt{2}} dx \leq \int_0^{\pi/2} \frac{dx}{\sqrt{1+\sin^3 x}} \leq \int_0^{\pi/2} dx$$

$$\therefore \frac{\pi}{2\sqrt{2}} \leq I \leq \frac{\pi}{2}$$

Q19 (1, 2, 3) - Multiple Correct

$$\text{We have, } \int_0^\infty e^{x^2} dx = \frac{\sqrt{\pi}}{2}$$

Now, check options.

$$(1) \int_0^\infty e^{-2x^2} dx$$

$$\text{Put } \sqrt{2}x = t \Rightarrow dx = \frac{dt}{\sqrt{2}}$$

$$\therefore \int_0^\infty e^{-2x^2} dx = \frac{1}{\sqrt{2}} \int_0^\infty e^{-t^2} dt$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{\pi}}{2} = \frac{\sqrt{\pi}}{2\sqrt{2}}$$

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$$(2) \int_0^\infty x e^{-x^2} dx$$

$$\begin{aligned} \text{Put } x^2 = t &\Rightarrow x dx = \frac{1}{2} dt \\ \therefore \int_0^\infty x e^{-x^2} dx &= \frac{1}{2} \int_0^\infty e^{-t} dt = \frac{1}{2} [-e^{-t}]_0^\infty \\ &= -\frac{1}{2} [e^{-\infty} - e^0] = -\frac{1}{2} [0 - 1] = \frac{1}{2} \end{aligned}$$

$$(3) \int_0^\infty x^2 e^{-x^2} dx$$

$$\begin{aligned} \text{Put } x^2 = t &\Rightarrow 2x dx = dt \\ \Rightarrow dx &= \frac{dt}{2\sqrt{t}} \\ \therefore \int_0^\infty x^2 e^{-x^2} dx &= \int_0^\infty \frac{t \cdot e^{-t}}{2\sqrt{t}} dt \\ &= \frac{1}{2} \int_0^\infty \sqrt{t} \cdot e^{-t} dt = \frac{1}{2} \int_0^\infty \sqrt{t} e^{-t} dt \\ &= \frac{1}{2} \left[\left[(-\sqrt{t} e^{-t}) \right]_0^\infty + \int_0^\infty \frac{1}{2\sqrt{t}} e^{-t} dt \right] \\ &= \frac{1}{2} \left[0 + \frac{1}{2} \int_0^\infty t^{-\frac{1}{2}} e^{-t} dt \right] = \frac{1}{4} \int_0^\infty t^{-\frac{1}{2}} \cdot e^{-t} dt \end{aligned}$$

$$\begin{aligned} \text{We know that, } \int_0^\infty e^{-x} x^{n-1} dx &= \Gamma n \\ \Rightarrow \frac{1}{4} \int_0^\infty t^{\frac{1}{2}-1} e^{-t} dt &= \frac{1}{4} \Gamma \frac{1}{2} = \frac{1}{4} \sqrt{\pi} = \frac{\sqrt{\pi}}{4} \end{aligned}$$

Q20 (2, 4) - Multiple Correct

$$\text{We have, } (6x + 1)x = 1 + \left[\cos \frac{\pi}{4} \right]$$

$$\Rightarrow 6x^2 + x - 1 = 0$$

$$\Rightarrow (2x + 1)(3x - 1) = 0$$

$$\Rightarrow \alpha = -\frac{1}{2} \text{ and } \beta = \frac{1}{3}$$

$$\text{Now, } \int_0^{2\alpha} \sin \left[\frac{\pi[x]}{2} \right] dx + \int_0^{3\beta} \cos(\pi[x]) dx$$

$$= \int_0^{-1} \sin \left(\frac{\pi[x]}{2} \right) + \int_0^1 \cos(\pi[x]) dx$$

$$= \int_0^1 \cos(\pi[x]) dx - \int_{-1}^0 \sin \left(\frac{\pi[x]}{2} \right) dx$$

$$= \int_0^1 \cos 0 dx - \int_{-1}^0 \sin \left(-\frac{\pi}{2} \right) dx$$

$$= [x]_0^1 + [x]_{-1}^0$$

$$= 1 + 0 - (-1)$$

$$= 1 + 1 = 2$$

$$\text{Now, } \alpha + \beta \neq 2$$

$$\therefore 2\alpha + 9\beta = -1 + 3 = 2$$

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Q21 (3, 4) - Multiple Correct

$$\text{Given, } f(x) = \int_1^x \frac{\log t}{1+t} dt$$

$$\text{Here, } f(x) + f\left(\frac{1}{x}\right) = \frac{\log^2 x}{2} = 2$$

$$\Rightarrow \log_e x = \pm 2$$

$$\therefore x = e^{\pm 2}$$

Q22 (3, 4) - Multiple Correct

$$f(x) = \int_{-1}^1 (1 - |t|) \cos xtdt$$

$$= 2 \int_0^1 (1 - |t|) \cos xtdt$$

$$= 2 \left[\left\{ (1 - t) \frac{\sin xt}{x} \right\}_0^1 - \int_0^1 (-1) \frac{\sin xt}{x} dt \right]$$

$$= 2 \left[0 + \frac{-\cos xt}{x^2} \right]_0^1 = \frac{2(1 - \cos x)}{x^2}$$

$$= -\frac{2}{x^2} [\cos x - 1]$$

$$\therefore \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{2(2 \sin^2 \frac{x}{2})}{x^2} = \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{2}}{\frac{x^2}{4}} = 1$$

$$\Rightarrow f(0) = \int_{-1}^1 (1 - |t|) dt$$

$$= 2 \int_0^1 (1 - t) dt = 2 \left[t - \frac{t^2}{2} \right]_0^1$$

$$= 2 \left(1 - \frac{1}{2} \right) = 2 \times \frac{1}{2} = 1$$

Hence, $f(x)$ is continuous at $x = 0$.

Q23 (1, 2, 3) - Multiple Correct

We have,

$$f[f(x)] = -x + 1$$

On replacing x by $f(x)$, we get

$$f[f\{f(x)\}] = -f(x) + 1$$

$$\Rightarrow f(1 - x) = -f(x) + 1$$

$$\Rightarrow f(x) + f(1 - x) = 1 \quad \dots (i)$$

$$\text{Now, } J = \int_0^1 f(x) dx = \int_0^1 f(1 - x) dx$$

$$\Rightarrow 2J = \int_0^1 [f(x) + f(1 - x)] dx$$

$$\Rightarrow 2J = \int_0^1 dx = 1$$

$$\therefore J = \frac{1}{2}$$

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$$\text{Let } K = \int_0^{\pi/2} \frac{\sin x}{(\sin x + \cos x)^3} dx \dots \text{(ii)}$$

$$K = \int_0^{\pi/2} \frac{\cos x}{(\cos x + \sin x)^3} dx \dots \text{(iii)}$$

From Eqs. (ii) and (iii), we get

$$2K = \int_0^{\pi/2} \frac{\sin x + \cos x}{(\sin x + \cos x)^3} dx$$

$$\Rightarrow 2K = \int_0^{\pi/2} \frac{1}{(\sin x + \cos x)^2} dx$$

$$= \int_0^{\pi/2} \frac{1}{1 + 2 \sin x \cos x} dx$$

$$= \int_0^{\pi/2} \frac{\sec^2 x}{1 + \tan^2 x + 2 \tan x} dx$$

$$\text{Put } 1 + \tan x = t \Rightarrow \sec^2 x dx = dt$$

$$\Rightarrow 2K = \int_1^\infty \frac{dt}{t^2} = \left[-\frac{1}{t} \right]_1^\infty = -[0 - 1]$$

$$\Rightarrow 2K = 1$$

$$\text{Then, } K = \frac{1}{2} = J$$

Q24 (3, 4) - Multiple Correct

$$\text{We have, } [f'(x)]^2 + [g(x)]^2 = 1$$

$$\text{and } f(x) + \int_0^x g(t) dt = \sin x (\cos x - \sin x)$$

On differentiating both sides, we get

$$f'(x) + g(x) = \cos 2x - \sin 2x$$

On squaring both sides, we get

$$[f'(x)]^2 + [g(x)]^2 + 2f'(x) \cdot g(x) = 1 - \sin 4x$$

$$\Rightarrow 2f'(x) \cdot g(x) = -\sin 4x$$

$$g(x) = -\frac{\sin 4x}{2f'(x)}, \text{ substituting in Eq. (i)}$$

$$f'(x) - \frac{\sin 4x}{2f'(x)} = \cos 2x - \sin 2x$$

On putting $f'(x) = t$

$$2t^2 - 2(\cos 2x - \sin 2x)t - \sin 4x = 0$$

$$\Rightarrow t = \frac{2(\cos 2x - \sin 2x) \pm \sqrt{4(1 - \sin 4x) + 8 \sin 4x}}{4}$$

$$\Rightarrow 2t = \cos 2x - \sin 2x \pm \sqrt{1 + 4 \sin 4x}$$

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On taking positive sign, we get

$$2t = \cos 2x - \sin 2x + \sqrt{1 + 4 \sin 4x}$$

$$= \cos 2x - \sin 2x + \cos 2x + \sin 2x$$

$$\Rightarrow t = \cos 2x$$

On taking negative sign, we get

$$t = -\sin 2x$$

Since, $f'(x) = \cos 2x$ or $f'(x) = -\sin 2x$

$$f(x) = \frac{1}{2} \sin 2x + C_1$$

$$\text{Or } f(x) = \frac{\cos 2x}{2} + C_2$$

$$f(0) = 0 \Rightarrow C_1 = 0 \text{ and } C_2 = \frac{-1}{2}$$

$$f(x) = \frac{1}{2} \sin 2x \text{ or } f(x) = \frac{\cos 2x - 1}{2}$$

If $f'(x) = \cos 2x$, then $g(x) = -\sin 2x$

If $f(x) = \frac{\cos 2x - 1}{2}$, then $g(x) = \cos 2x$

$\therefore f(x) = -\sin^2 x$ and $g(x) = \cos 2x$

Q25 (2) - Multiple Correct

$$\text{Let } I = \int_{-2008}^{2008} \frac{f'(x) + f'(-x)}{(2008)^x + 1} dx \dots (i)$$

$$\text{Put } x = -\theta \Rightarrow dx = -d\theta$$

$$\therefore I = - \int_{2008}^{-2008} \frac{f'(-\theta) + f'(\theta)}{(2008)^{-\theta} + 1} d\theta$$

$$= \int_{-2008}^{2008} \frac{(2008)^{\theta} [f'(\theta) + f'(-\theta)]}{1 + (2008)^{\theta}} d\theta \dots (ii)$$

On adding Eqs. (i) and (ii), we get

$$2I = \int_{-2008}^{2008} [f'(\theta) + f'(-\theta)] d\theta$$

$$\Rightarrow 2I = 2 \int_0^{2008} [f'(\theta) + f'(-\theta)] d\theta$$

$$\therefore [f(\theta) - f(-\theta)]_0^{2008}$$

$$= f(2008) - f(-2008)$$

Q26 (3) - Multiple Correct

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Given, $I = \int_0^{1/2} \frac{dx}{\sqrt{1-x^{2n}}}$, since $n \geq 1$ and $0 \leq x \leq \frac{1}{2}$

$$\therefore \sqrt{1-x^2} \leq \sqrt{1-x^{2n}} \leq 1$$

$$\frac{1}{\sqrt{1-x^2}} \geq \frac{1}{\sqrt{1-x^{2n}}} \geq 1$$

$$\Rightarrow \int_0^{1/2} 1 dx \leq 1 \leq \int_0^{1/2} \frac{dx}{\sqrt{1-x^2}}$$

$$\therefore \frac{1}{2} \leq 1 \leq \frac{\pi}{6}$$

Q27 (3) - Multiple Correct

We have, $f(4-x) = f(4+x)$

On replacing x by $4-x$, we get

$$f(4-4+x) = f(4+4-x)$$

$$f(x) = f(8-x) = f(x+8)$$

Thus, $f(x)$ is periodic function with period 8.

$$\text{Now, } \int_0^{200} f(x) dx = \int_0^{8 \times 25} f(x) dx$$

$$= 25 \int_0^8 f(x) dx = 25 \times 5 = 125$$

Q28 (2) - Multiple Correct

We have, $\int_{19}^{37} (\{x\}^2 + 3 \sin 2\pi x) dx$

Since, given integrand is periodic function with period 1 $\therefore (37-19) \int_0^1 [(\{x\})^2 + 3 \sin 2\pi x] dx$

$$= 18 \int_0^1 (x^2 + 3 \sin 2\pi x) dx$$

$$= 18 \left[\frac{x^3}{3} - \frac{3}{2\pi} \cos 2\pi x \right]_0^1$$

$$= 18 \left[\left(\frac{1}{3} - \frac{3}{2\pi} \right) + \frac{3}{2\pi} \right]$$

$$= \frac{18}{3} = 6$$

Q29 (1) - Multiple Correct

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We have,

$$\begin{aligned}
 I &= \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx \\
 &= \int_0^1 \frac{x^4}{1+x^2} (1-4x+6x^2-4x^3+x^4) dx \\
 &= \int_0^1 \frac{x^4-4x^5+6x^6-4x^7+x^8}{1+x^2} dx \\
 &= \int_0^1 \left(x^6-4x^5+5x^4-4x^2+4-\frac{4}{1+x^2} \right) dx \\
 &= \left[\frac{x^7}{7} - \frac{2x^6}{3} + x^5 - \frac{4x^3}{3} + 4x \right]_0^1 - \pi \\
 &= \frac{1}{7} - \frac{2}{3} + 1 - \frac{4}{3} + 4 - \pi = \frac{22}{7} - \pi
 \end{aligned}$$

Q30 (1, 4) - Multiple Correct

$$\begin{aligned}
 \text{Given, } S_n &= \sum_{k=0}^n \frac{n}{n^2+kn+k^2} \\
 &= \sum_{k=0}^n \frac{1}{n} \cdot \left(\frac{1}{1+\frac{k}{n}+\frac{k^2}{n^2}} \right) < \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{1}{n} \left(\frac{1}{1+\frac{k}{n}+\left(\frac{k}{n}\right)^2} \right) \\
 &= \int_0^1 \frac{1}{1+x+x^2} dx \\
 &= \left[\frac{2}{\sqrt{3}} \tan^{-1} \left\{ \frac{2}{\sqrt{3}} \left(x + \frac{1}{2} \right) \right\} \right]_0^1 \\
 &= \frac{2}{\sqrt{3}} \cdot \left(\frac{\pi}{3} - \frac{\pi}{6} \right) = \frac{\pi}{3\sqrt{3}}
 \end{aligned}$$

$$\text{i.e. } S_n < \frac{\pi}{3\sqrt{3}}$$

$$\text{Similarly, } T_n > \frac{\pi}{3\sqrt{3}}$$

Q31 (2, 4) - Multiple Correct

Here,

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{(1^a+2^a+\dots+n^a)}{(n+1)^{a-1}\{(na+1)+(na+2)+\dots+(na+n)\}} &= \frac{1}{60} \\
 \Rightarrow \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^a}{(n+1)^{a-1} \left[n^2 a + \frac{n(n+1)}{2} \right]} &= \frac{1}{60} \\
 \Rightarrow \lim_{n \rightarrow \infty} \frac{2 \sum_{r=1}^n \left(\frac{r}{n} \right)^a}{\left(1 + \frac{1}{n} \right)^{a-1} (2na+n+1)} &= \frac{1}{60} \\
 \Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n} \left[2 \sum_{r=1}^n \left(\frac{r}{n} \right)^a \right] \cdot \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n} \right)^{a-1} \left(2a + 1 + \frac{1}{n} \right)} &= \frac{1}{60}
 \end{aligned}$$

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$$\Rightarrow 2 \int_0^1 (x^a) dx \cdot \frac{1}{(2a+1)} = \frac{1}{60}$$

$$\Rightarrow \frac{2 \cdot [x^{a+1}]_0^1}{(2a+1)(a+1)} = \frac{1}{60}$$

$$\Rightarrow \frac{2}{(2a+1)(a+1)} = \frac{1}{60}$$

$$\therefore \frac{2}{(2a+1)(a+1)} = \frac{1}{60}$$

$$\Rightarrow (2a+1)(a+1) = 120$$

$$\Rightarrow 2a^2 + 3a + 1 - 120 = 0$$

$$\Rightarrow 2a^2 + 3a - 119 = 0$$

$$\Rightarrow (2a+17)(a-7) = 0$$

$$\therefore a = 7, -\frac{17}{2}$$

Q32 (1) - Paragraph 1

$$\text{We have, } f(x) = \int_0^1 e^{x+t} f(t) dt$$

$$\Rightarrow f(x) = e^x \int_0^1 e^t f(t) dt \quad \text{or} \quad f(x) = A \cdot e^x$$

$$\text{where, } A = \int_0^1 e^t f(t) dt$$

$$f(t) = Ae^t$$

$$\text{where, } A = \int_0^1 e^t f(t) dt$$

$$A = \int_0^1 e^t \cdot A \cdot e^t dt = A \int_0^1 e^{2t} dt$$

$$\text{which implies either } A = 0 \text{ or } \int_0^1 e^{2t} dt = 0$$

$$\text{But } \int_0^1 e^{2t} dt \neq 0$$

$$\text{Hence, } f(x) = 0 \Rightarrow f(1) = 0$$

Q33 (1) - Paragraph 1

$$\text{Given, } g(x) = e^x \int_0^1 e^t g(t) dt + x$$

$$\Rightarrow g(x) = Be^x + x$$

$$\text{or } g(t) = Be^t + t, \text{ where } g(t) = Be^t + t$$

$$B = \int_0^1 e^t (Be^t + t) dt$$

$$\Rightarrow B = B \int_0^1 e^{2t} dt + \int_0^1 e^t \cdot t dt$$

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But $\int_0^1 e^{2t} dt = \frac{1}{2}(e^2 - 1)$ and $\int_0^1 te^t dt = 1$

$\therefore B = \frac{B}{2}(e^2 - 1) + 1$

$\Rightarrow 2B = B(e^2 - 1) + 2$

$\Rightarrow 3B = Be^2 + 2 \Rightarrow B = \frac{2}{3-e^2}$

$\Rightarrow g(x) = \left(\frac{2}{3-e^2}\right)e^x + x$

$\Rightarrow g(0) = \frac{2}{3-e^2}$ and $f(0) = 0$

$\Rightarrow g(0) - f(0) = \frac{2}{3-e^2}$

Q34 (3) - Paragraph 2

$D(t) = \frac{10+0-11}{\sqrt{a^2e^{2t}+b^2e^{-2t}}} = \frac{1}{\sqrt{a^2e^{2t}+b^2e^{-2t}}}$

$I = \int_0^1 \frac{dt}{[D(t)]^2} = \int_0^1 (a^2e^{2t} + b^2e^{-2t}) dt$

$= \left[a^2 \frac{1}{2} e^{2t} - \frac{b^2}{2} e^{-2t} \right]_0^1 = \frac{1}{2} [(a^2e^2 - b^2e^{-2}) - (a^2 - b^2)]$

$= \frac{1}{2} [a^2(e^2 - 1) - b^2(e^{-2} - 1)] = \frac{1}{2} \left[a^2(e^2 - 1) - b^2 \frac{(1-e^2)}{e^2} \right]$

$= \frac{1}{2} (e^2 - 1) \left[a^2 + \frac{b^2}{e^2} \right] = \frac{1}{2} (e^2 - 1) \left(a^2 + \frac{b^2}{e^2} \right)$

Q35 (4) - Paragraph 2

We have, $1 = \frac{e^2-1}{2} \left(a^2 + \frac{b^2}{e^2} \right)$

We have, $ab = 1 \Rightarrow b = \frac{1}{a}$

$\therefore I = \frac{e^2-1}{2} \left(a^2 + \frac{1}{a^2e^2} \right)$

$= \frac{e^2-1}{2} \left[\left(a - \frac{1}{ae} \right)^2 + \frac{2}{e} \right]$

is minimum, if $a = \frac{1}{ae}$.

$\therefore a^2 = \frac{1}{e} \Rightarrow \frac{1}{b^2} = \frac{1}{e}$

$\Rightarrow b = \sqrt{e}$

Q36 (4) - Paragraph 3

Given, $S_n = \int_0^{\pi/2} \frac{\sin(2n-1)x}{\sin x} dx$

Then, $S_{n+1} = \int_0^{\pi/2} \frac{\sin(2n+1)x}{\sin x} dx$

$\therefore S_{n+1} - S_n = \int_0^{\pi/2} \frac{\{\sin(2n+1)x - \sin(2n-1)x\}}{\sin x} dx$

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$$= 2 \int_0^{\pi/2} \frac{\sin x \cdot \cos(2n)x}{\sin x} dx$$

$$= 2 \int_0^{\pi/2} \cos(2n)x dx$$

$$= \frac{2 \cdot [\sin(2nx)]_0^{\pi/2}}{2n} = \frac{1}{n} \cdot (0) = 0$$

Q37 (2) - Paragraph 3

$$\text{Given, } V_n = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x} dx$$

$$\text{Then, } V_{n+1} = \int_0^{\pi/2} \frac{\sin^2(n+1)x}{\sin^2 x} dx$$

$$\therefore V_{n+1} - V_n = \int_0^{\pi/2} \frac{\sin^2(n+1)x - \sin^2 nx}{\sin^2 x} dx$$

$$= \int_0^{\pi/2} \frac{\sin(2n+1)x \cdot \sin x}{\sin^2 x} dx$$

$$\therefore V_{n+1} - V_n = S_{n+1}$$

Q38 (1) - Paragraph 4

$$\int_0^1 f(x) dx = \int_0^1 \frac{\sin x - x \cos x}{x^2} dx$$

$$= \int_0^1 \frac{\sin x}{x^2} dx - \int_0^1 \frac{\cos x}{x} dx$$

$$= \left[\sin x \left(-\frac{1}{x} \right) \right]_0^1 + \int_0^1 \cos x \left(\frac{1}{x} \right) dx - \int_0^1 \frac{\cos x}{x} dx$$

$$= - \left[\frac{\sin x}{x} \right]_0^1$$

$$= 1 - \sin(1)$$

Q39 (2) - Paragraph 4

$$\lim_{t \rightarrow 0} \frac{\int_0^t f(x) dx}{t^2} = \lim_{t \rightarrow 0} \int_0^t \frac{\sin x - x \cos x}{\frac{x^2}{t^2}} dx$$

Using L' Hospital's rule, we have

$$\lim_{t \rightarrow 0} \frac{\sin t - t \cos t}{t^2 \cdot 2t} = \lim_{t \rightarrow 0} \frac{\cos t (\tan t - t)}{2t^3}$$

$$= \frac{1}{2} \lim_{t \rightarrow 0} \frac{\cos^2 t - 1}{3t^2} = \frac{1}{6}$$

Q40 (4) - Integer Type

Since, I is independent of x , so $\frac{dI}{dx} = 0$.

$$\Rightarrow f(xy) \cdot y - f(x) = 0$$

$$\Rightarrow f(xy) \cdot y = f(x)$$

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On putting, $y = \frac{2}{x} \Rightarrow f(2) \left(\frac{2}{x} \right) = f(x)$

$$\Rightarrow f(x) = \frac{4}{x} \quad [\because f(2) = 2]$$

$$\therefore \int_1^e \frac{4}{x} dx = 4 \log e = 4$$

Q41 (4) - Integer Type

$$I = \int_{k\pi}^{(k+1)\pi} \frac{|\sin 2x|}{|\sin x| + |\cos x|} dx$$

Given integrand is periodic with period $\frac{\pi}{2}$

$$I = \int_0^\pi \frac{|\sin 2x|}{|\sin x| + |\cos x|} dx$$

$$= 2 \int_0^{\pi/2} \frac{|\sin 2x|}{|\sin x| + |\cos x|} dx$$

$$\Rightarrow I = 2 \cdot 2 \int_0^{\pi/4} \frac{|\sin 2x|}{|\sin x| + |\cos x|} dx$$

$$= 4 \int_0^{\pi/4} \frac{|\sin 2x|}{|\sin x| + |\cos x|} dx$$

$$\Rightarrow I = 4 \int_0^{\pi/4} \frac{\sin 2x}{\sin x + \cos x} dx \quad \dots (i)$$

$$\text{and } J = \int_0^{\pi/4} \frac{1}{\sin x + \cos x} dx \quad \dots (ii)$$

$$\Rightarrow I + 4J = 4 \int_0^{\pi/4} \frac{1 + \sin 2x}{\sin x + \cos x} dx$$

$$= 4 \int_0^{\pi/4} (\sin x + \cos x) dx$$

$$= 4[-\cos x + \sin x]_0^{\pi/4}$$

$$= 4[0 - (-1)] = 4$$

Q42 (1) - Integer Type

We have, $f(x + 13) + f(x + 630) = 0$

On replacing x by $x - 13$, we get

$$f(x) + f(x + 617) = 0 \dots (i)$$

Again, on replacing x by $x + 617$, we get

$$f(x + 617) + f(x + 1234) = 0 \dots (ii)$$

On subtracting Eq. (i) from Eq. (ii), we get

$$f(x + 1234) = f(x)$$

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$\because f(x)$ is periodic with period $T = 1234$

$$\text{Now, } \sum_{p=1}^{777} \int_p^{p+T} f(x) dx = \lambda \int_0^T f(x) dx$$

$$\Rightarrow \sum_{p=1}^{777} \int_0^T f(x) dx = \lambda \int_0^T f(x) dx$$

$$\Rightarrow \int_0^T f(x) dx \times 777 = \lambda \int_0^T f(x) dx$$

$$\therefore \lambda = 777$$

$$\text{Then, } \frac{\lambda+T}{2011} = \frac{1234+777}{2011} = \frac{2011}{2011} = 1$$

Q43 (5) - Integer Type

We have, $f(2-x) = f(2+x)$

On replacing x by $2-x$, we get

$$f(2-2+x) = f(2+2-x)$$

$$f(x) = f(4-x) = f(x+4)$$

Since, $f(x)$ is periodic with period 4

$$\text{We have, } \int_0^2 f(x) dx = 5$$

$$\Rightarrow \frac{1}{25} \int_0^{50} f(x) dx = \frac{1}{25} \times \frac{25}{2} \int_0^4 f(x) dx$$

$$= \frac{1}{2} \int_0^4 f(x) dx = \int_0^2 f(x) dx$$

$$= 5$$

Q44 (1) - Integer Type

$$\text{We have, } f(x) = \int_a^x [f(x)]^{-2} dx$$

$$f'(x) = [f(x)]^{-2} \cdot 1$$

$$\therefore [f(x)]^2 f'(x) = 1$$

Q45 (1) - Integer Type

$$\text{Let } I = \int_0^x \left(x - [x] - \frac{1}{2} \right) dx$$

$$= \int_0^{[x]+\{x\}} \left(x - [x] - \frac{1}{2} \right) dx$$

$$= \int_0^{[x]} \left(\{x\} - \frac{1}{2} \right) dx + \int_{[x]}^{[x]+\{x\}} \left(\{x\} - \frac{1}{2} \right) dx$$

$$= [x] \int_0^1 \left(\{x\} - \frac{1}{2} \right) dx + \int_0^{\{x\}} \left(\{x\} - \frac{1}{2} \right) dx$$

$$= [x] \int_0^1 \left(x - \frac{1}{2} \right) dx + \int_0^{\{x\}} \left(x - \frac{1}{2} \right) dx$$

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$$= [x] \left(\frac{x^2}{2} - \frac{x}{2} \right)_0^1 + \left(\frac{x^2}{2} - \frac{1}{2}x \right)_{\{x\}}^{\{x\}}$$

$$= 0 + \frac{\{x\}(\{x\}-1)}{2} = \frac{\{x\}(\{x\}-1)}{2}$$

$$\text{On comparing, } \frac{f(x) \cdot g(x)}{2} = \frac{\{x\}(\{x\}-1)}{2}$$

$$\therefore |f(x) - g(x)| = 1$$

Q46 (1) - Integer Type

$$\text{Let } I(\alpha) = \int_0^1 \frac{x^\alpha - 1}{\log x} dx$$

$$\text{Then, } \frac{d[I(\alpha)]}{d\alpha} = \int_0^1 \frac{x^\alpha \cdot \log x}{\log x} dx = \int_0^1 x^\alpha dx$$

$$= \left[\frac{x^{\alpha+1}}{\alpha+1} \right]_0^1 = \frac{1}{\alpha+1}$$

$$\Rightarrow \frac{d[I(\alpha)]}{d\alpha} = \frac{1}{\alpha+1}$$

On integrating w.r.t. α , we get

$$I(\alpha) = \log |\alpha + 1| + C \dots (i)$$

$$\text{If } \alpha = 0, \text{ then } I(0) = \int_0^1 \frac{x^0 - 1}{\log x} dx = 0$$

On putting $\alpha = 0$ in Eq. (i), we get

$$0 = \log(1) + C \Rightarrow C = 0$$

$$\Rightarrow k = 1$$

Q47 (3) - Integer Type

We have,

$$\lim_{n \rightarrow \infty} \frac{\pi}{6n} \left[\sec^2 \left(\frac{\pi}{6n} \right) + \sec^2 \left(2 \cdot \frac{\pi}{6n} \right) + \dots + \sec^2 \left((n-1) \frac{\pi}{6n} \right) + \frac{4}{3} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{\pi}{6} \cdot \frac{1}{n} \cdot \sum_{r=0}^n \sec^2 \left(\frac{r \cdot \pi}{6n} \right)$$

$$= \frac{\pi}{6} \cdot \int_0^1 \sec^2 \left(\frac{\pi x}{6} \right) dx = \frac{\pi}{6} \cdot \left[\frac{\tan \left(\frac{\pi x}{6} \right)}{\frac{\pi}{6}} \right]_0^1$$

$$= \tan \frac{\pi}{6} - 0 = \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{k}}$$

$$\therefore k = 3$$

Q48 (8) - Integer Type

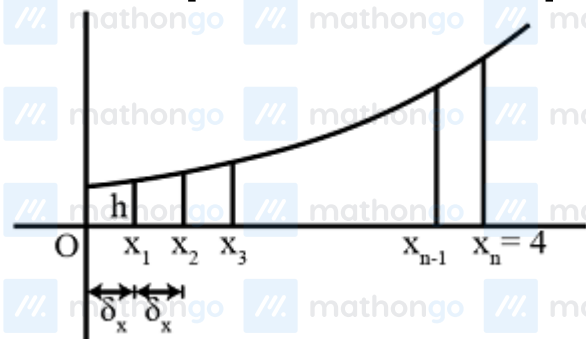
Hints & Solutions

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We have,

$$\lim_{\delta x \rightarrow 0} \delta x (x_1 + x_2 + x_3 + \dots + x_n)$$

$$= \lim_{n \rightarrow \infty} \frac{4}{n} \left[\frac{4}{n} + \frac{8}{n} + \frac{12}{n} + \dots + 4 \cdot \frac{n}{n} \right] \quad \left[\because \delta x = \frac{4}{n} \right]$$



$$= \lim_{n \rightarrow \infty} \frac{16}{n^2} (1 + 2 + 3 + \dots + n)$$

$$= \lim_{n \rightarrow \infty} \frac{16}{n^2} \cdot \frac{n(n+1)}{2} = 8$$

Q49 (2) - Integer Type

$$\text{Let } I = \int_0^\infty x^{2n} \cdot x \cdot e^{-x^2} dx$$

$$\text{Put } x^2 = t \Rightarrow 2x dx = dt$$

$$\therefore I = \frac{1}{2} \int_0^\infty t^n \cdot e^{-t} dt = \frac{1}{2} \int_0^\infty e^{-t} \cdot t^{(n+1)-1} dt$$

$$= \frac{1}{2} \Gamma(n+1) = \frac{n!}{2} \quad [\text{as } \Gamma(n+1) = n!, n \in \mathbb{N}]$$

$$\therefore k = 2$$

Q50 (4) - Integer Type

$$\text{Let } I = \int_0^{400\pi} \sqrt{1 - \cos 2x} dx$$

$$= \int_0^{400\pi} \sqrt{2 \cdot \sin^2 x} dx$$

$$= 400\sqrt{2} \int_0^{400\pi} |\sin x| dx$$

$$= 400\sqrt{2} \int_0^\pi |\sin x| dx \quad [\because \text{period is } \pi]$$

$$= 400\sqrt{2} \int_0^\pi \sin x dx$$

$$= -400\sqrt{2} [\cos x]_0^\pi$$

$$= -400\sqrt{2} (-1 - 1)$$

$$= 800\sqrt{2}$$

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But $k \cdot 200\sqrt{2} = 800\sqrt{2}$

$\Rightarrow k = 4$

Q51 (2) - Integer Type

Given, $f(x) = \int_0^{x^2} \frac{t^2 - 5t + 4}{e^t + 2} dt$

$\Rightarrow f'(x) = \frac{x^4 - 5x^2 + 4}{e^{x^2} + 2} \cdot 2x$

$f'(x) = 0$

$\Rightarrow x^5 - 5x^3 + 4x = 0$

$\Rightarrow x(x^4 - 5x^2 + 4) = 0$

$\Rightarrow x = 0 \text{ and } x^4 - 5x^2 + 4 = 0$

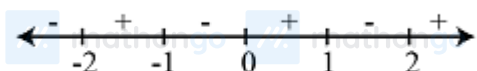
$\Rightarrow x = 0 \text{ and } x^4 - 4x^2 - x^2 + 4 = 0$

$\Rightarrow x^2(x^2 - 4) - 1(x^2 - 4) = 0$

$\Rightarrow (x^2 - 4)(x^2 - 1) = 0$

$\Rightarrow (x - 2)(x + 2)(x - 1)(x + 1) = 0$

$\Rightarrow x = \pm 2, \pm 1, 0$



$\therefore f'(x)$ changes sign from positive to negative at $x = -1$ and $x = 1$

$\therefore$ Points of local maxima is 2.